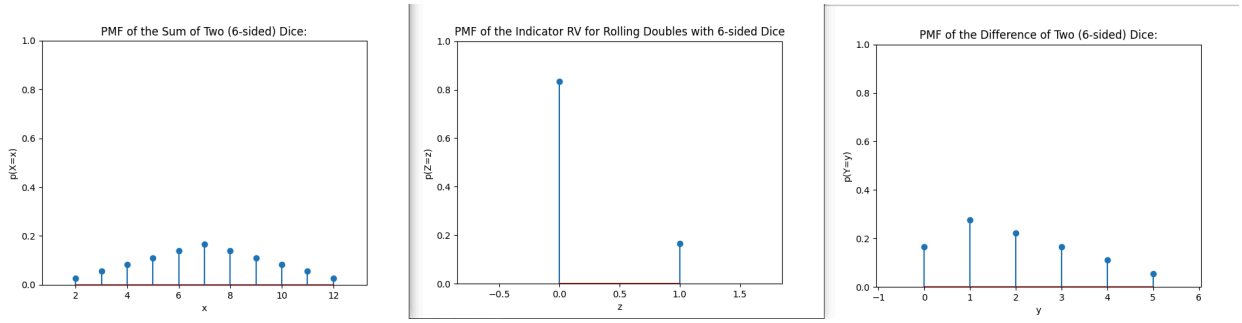
Coding Assignment 1
Austin Erickson

Terminal output:
1.4:



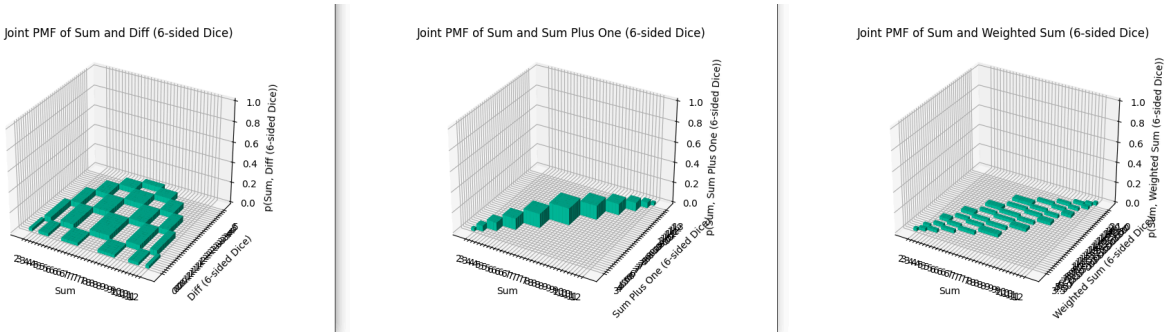
Expected Value of the Sum of Two (6-sided) Dice: 7.0
Variance of the Sum of Two (6-sided) Dice: 5.833333333333334
Expected Value of the Diff. of Two (6-sided) Dice: 1.9444444444444446
Variance of the Difference of Two (6-sided) Dice: 2.052469135802469
Expected Value of the Indicator RV for Rolling Doubles (6-sided Dice): 0.16666666666666669

(base) austiner99@radlab15:~/classes/localization_and_mapping_f26$ pytest coding_ass_1/lab1/test/test_part1.py
========== test session starts ==========
platform linux -- Python 3.8.10, pytest-8.3.5, pluggy-1.5.0
rootdir: /home/austiner99/classes/localization_and_mapping_f26/coding_ass_1/lab1
configfile: pytest.ini
collected 15 items

coding_ass_1/lab1/test/test_part1.py .....
[100%]

========== 15 passed in 0.30s ==========

2.2.3:



{(2, 0): 0.027777777777777776, (3, 1): 0.05555555555555555, (4, 2): 0.05555555555555555, (5, 3): 0.05555555555555555, (6, 4): 0.05555555555555555, (7, 5): 0.05555555555555555, (4, 0): 0.027777777777777776, (5, 1): 0.05555555555555555, (6, 2): 0.05555555555555555, (7, 3): 0.05555555555555555, (8, 4): 0.05555555555555555, (6, 0): 0.027777777777777776, (7, 1): 0.05555555555555555, (8, 2): 0.05555555555555555, (9, 3): 0.05555555555555555, (8, 0): 0.027777777777777776, (9, 1): 0.05555555555555555, (10, 2): 0.05555555555555555, (10, 0): 0.027777777777777776, (11, 1): 0.05555555555555555, (12, 0): 0.027777777777777776}
Expected value (Joint of sum/diff) with 6-sided Dice: [7. 1.94444444]
Covariance (Joint of sum/diff) with 6-sided Dice: [[5.83333333 0.]
[0. 2.05246914]]
Sum and difference are uncorrelated because their covariance is zero.
{(2, 3): 0.027777777777777776, (3, 4): 0.05555555555555555, (4, 5): 0.08333333333333333, (5, 6): 0.111111111111111, (6, 7): 0.1388888888888889, (7, 8): 0.16666666666666669, (8, 9): 0.1388888888888889,(9, 10): 0.1111111111111111, (10, 11): 0.08333333333333333, (11, 12): 0.05555555555555555, (12, 13): 0.027777777777777776}
Expected value (Joint of sum/sum_plus_one) with 6-sided Dice: [7. 8.]
Covariance (Joint of sum/sum_plus_one) with 6-sided Dice: [[5.83333333 5.83333333]
[5.83333333 5.83333333]]
Sum and sum plus one are positively correlated.
{(2, 3.5): 0.027777777777777776, (3, 6.0): 0.027777777777777776, (4, 8.5): 0.027777777777777776, (5, 11.0): 0.027777777777777776, (6, 13.5): 0.027777777777777776, (7, 16.0): 0.027777777777777776, (3, 4.5): 0.027777777777777776, (4, 7.0): 0.027777777777777776, (5, 9.5): 0.027777777777777776, (6, 12.0): 0.027777777777777776, (7, 14.5): 0.027777777777777776, (8, 17.0): 0.027777777777777776, (4, 5.5): 0.027777777777777776, (5, 8.0): 0.027777777777777776, (6, 10.5): 0.027777777777777776, (7, 13.0): 0.027777777777777776, (8, 15.5): 0.027777777777777776, (9, 18.0): 0.027777777777777776, (5, 6.5): 0.027777777777777776, (6, 9.0): 0.027777777777777776, (7, 11.5): 0.027777777777777776, (8, 14.0): 0.027777777777777776, (9, 16.5): 0.027777777777777776, (10, 19.0): 0.027777777777777776, (6, 7.5): 0.027777777777777776, (7, 10.0): 0.027777777777777776, (8, 12.5): 0.027777777777777776, (9, 15.0): 0.027777777777777776, (10, 17.5): 0.027777777777777776, (11, 20.0): 0.027777777777777776, (7, 8.5): 0.027777777777777776, (8, 11.0): 0.027777777777777776, (9, 13.5): 0.027777777777777776, (10, 16.0): 0.027777777777777776, (11, 18.5): 0.027777777777777776, (12, 21.0): 0.027777777777777776}

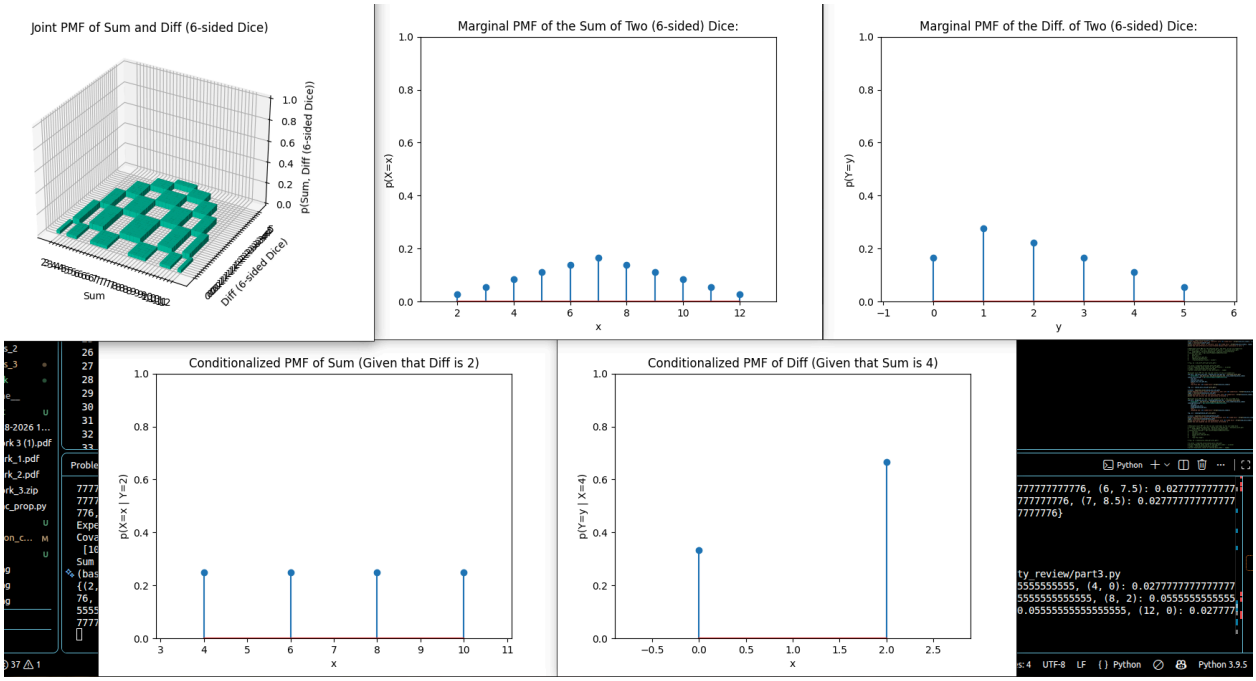
Expected value (Joint of sum/weightedsum) with 6-sided Dice: [7. 12.25]
Covariance (Joint of sum/weightedsum) with 6-sided Dice: [[5.83333333 10.20833333]
 [10.20833333 21.14583333]]
Sum and weighted sum are positively correlated.

base) austiner99@radlab15:~/classes/localization_and_mapping_f26$ pytest coding_ass_1/lab1/test/test_part2.py
========== test session starts ==========
platform linux -- Python 3.8.10, pytest-8.3.5, pluggy-1.5.0
rootdir: /home/austiner99/classes/localization_and_mapping_f26/coding_ass_1/lab1
configfile: pytest.ini
collected 17 items

coding_ass_1/lab1/test/test_part2.py .....
[100%]

========== 17 passed in 0.28s ==========

3.2.1:



(base) austiner99@radlab15:~/classes/localization_and_mapping_f26$ pytest coding_ass_1/lab1/test/test_part3.py
========== test session starts ==========
platform linux -- Python 3.8.10, pytest-8.3.5, pluggy-1.5.0
rootdir: /home/austiner99/classes/localization_and_mapping_f26/coding_ass_1/lab1
configfile: pytest.ini
collected 8 items

coding_ass_1/lab1/test/test_part3.py ....
[100%]

========== 8 passed in 0.27s ==========

Question Answers:

1.4: Because there are more combinations for the number 7 than any other number (6 and 1, 2 and 4, etc.) that number appears more, assuming uniform distribution of the numbers.

2.2.3: I believe that the only correlated joint distributions are “sum and sum plus one” and “sum and weighted sum” since covariance is present (as can be seen in the distributions), and it makes sense since both sum and sum plus one are derived from the sum.

3.2.1: The marginal PMF’s have similar shapes to the PMF’s generated in part 1.

The conditional PMFs output makes sense, as a sum of 4 means (achievable only through 1 and 3 or 2 and 2), results in only two possible diffs, with one of those able to occur twice as often. Same can be said with a diff of 2, achievable only with numbers 2 away from each other, meaning the sum will scale at an even level with the other sums.